
♦ **Chunk 1: Operating System Definition**

Definition:

An Operating System (OS) is system software that manages hardware and software resources and provides services to programs.

Explanation:

It acts as a bridge between user and hardware. Without OS, users must interact directly with hardware, which is complex.

Key Points:

- Interface between user and hardware
- Manages CPU, memory, files, devices
- Provides execution environment

Example:

Windows, Linux, macOS

Interview Tip:

Always mention “resource management + interface” in definition.

♦ **Chunk 2: Functions of OS**

Definition:

Core responsibilities performed by the OS.

Explanation:

OS ensures efficient and safe use of system resources.

Key Points:

- Process management
- Memory management
- File system management
- Device management
- Security

Example:

OS allocates CPU to running apps.

Interview Tip:

List at least 4 functions when asked.

♦ Chunk 3: Types of Operating Systems**Definition:**

Different categories based on functionality.

Explanation:

Each type is designed for specific use cases.

Key Points:

- Batch OS
- Time-sharing OS
- Distributed OS
- Real-time OS
- Multiprocessing OS

Example:

Real-time OS in embedded systems

Interview Tip:

RTOS = strict timing constraints.

♦ Chunk 4: Process**Definition:**

A process is a program in execution.

Explanation:

It includes code, data, stack, registers, and resources.

Key Points:

- Active entity
- Has state
- Requires resources

Example:

Running Chrome browser

Interview Tip:

Program = passive, Process = active.

♦ Chunk 5: Process States**Definition:**

Different stages of a process lifecycle.

Explanation:

Processes move between states during execution.

Key Points:

- New
- Ready
- Running
- Waiting
- Terminated

Example:

Waiting state → I/O operation

Interview Tip:

Draw state diagram if asked.

♦ Chunk 6: Process Scheduling**Definition:**

Selection of process for CPU execution.

Explanation:

Scheduler decides which process runs next.

Key Points:

- CPU utilization
- Throughput
- Turnaround time
- Waiting time

Example:

OS selects process from ready queue

Interview Tip:

Focus on performance metrics.

◆ Chunk 7: Scheduling Algorithms**Definition:**

Methods used to schedule processes.

Explanation:

Each algorithm has pros and cons.

Key Points:

- FCFS
- SJF
- Round Robin
- Priority Scheduling

Example:

Round Robin → time slice based

Interview Tip:

SJF gives minimum average waiting time.

◆ Chunk 8: Threads**Definition:**

Smallest unit of CPU execution.

Explanation:

Threads share process resources.

Key Points:

- Lightweight
- Faster than processes
- Shared memory

Example:

Multiple threads in a browser tab

Interview Tip:

Thread vs Process is very common question.

♦ Chunk 9: Process vs Thread**Definition:**

Comparison between process and thread.

Explanation:

Processes are independent, threads are not.

Key Points:

- Process → heavy
- Thread → lightweight
- Process → separate memory
- Thread → shared memory

Example:

Multiple threads inside one process

Interview Tip:

Mention memory sharing difference.

♦ Chunk 10: Inter-Process Communication (IPC)**Definition:**

Mechanism for processes to communicate.

Explanation:

Processes need IPC for coordination.

Key Points:

- Shared memory
- Message passing

Example:

Pipes in Linux

Interview Tip:

Shared memory is faster than message passing.

◆ Chunk 11: Deadlock

Definition:

A state where processes wait forever.

Explanation:

Occurs due to resource contention.

Key Points:

- Mutual exclusion
- Hold and wait
- No preemption
- Circular wait

Example:

Two processes waiting on each other

Interview Tip:

All 4 conditions must occur.

◆ Chunk 12: Deadlock Handling

Definition:

Methods to manage deadlocks.

Explanation:

OS prevents or resolves deadlocks.

Key Points:

- Prevention
- Avoidance
- Detection
- Recovery

Example:

Banker's Algorithm

Interview Tip:

Avoidance = safe state concept.

◆ **Chunk 13: Memory Management**

Definition:

Managing main memory efficiently.

Explanation:

OS allocates and frees memory.

Key Points:

- Allocation
- Deallocation
- Protection

Example:

RAM allocation to processes

Interview Tip:

Fragmentation is important concept.

◆ **Chunk 14: Paging**

Definition:

Memory management using fixed-size blocks.

Explanation:

Logical memory divided into pages.

Key Points:

- Page table
- No external fragmentation

Example:

Virtual memory systems

Interview Tip:

Internal fragmentation may occur.

◆ **Chunk 15: Segmentation**

Definition:

Memory divided into variable-sized segments.

Explanation:

Based on logical divisions.

Key Points:

- Code, data, stack segments
- External fragmentation

Example:

Separate code and data segments

Interview Tip:

Segmentation is user-visible.

◆ Chunk 16: Virtual Memory**Definition:**

Technique to use disk as memory.

Explanation:

Allows execution of large programs.

Key Points:

- Demand paging
- Page fault

Example:

Swapping pages from disk

Interview Tip:

Virtual memory improves multitasking.

◆ Chunk 17: File System**Definition:**

Structure for storing data.

Explanation:

OS organizes files and directories.

Key Points:

- File creation
- Deletion

- Access control

Example:

NTFS, EXT4

Interview Tip:

Know file allocation methods.

◆ **Chunk 18: Disk Scheduling**

Definition:

Order of disk access requests.

Explanation:

Improves performance.

Key Points:

- FCFS
- SSTF
- SCAN

Example:

Elevator algorithm (SCAN)

Interview Tip:

SCAN reduces seek time.

◆ **Chunk 19: System Calls**

Definition:

Interface between user and OS.

Explanation:

Programs request OS services.

Key Points:

- Process control
- File management
- Device control

Example:

open(), read()

Interview Tip:

System calls = OS API.

♦ Chunk 20: Kernel**Definition:**

Core part of OS.

Explanation:

Handles low-level operations.

Key Points:

- Process scheduling
- Memory management

Example:

Linux kernel

Interview Tip:

Kernel runs in privileged mode.

♦ Chunk 21: User Mode vs Kernel Mode**Definition:**

Two execution modes.

Explanation:

Ensures security.

Key Points:

- User mode → limited access
- Kernel mode → full access

Example:

System call switches mode

Interview Tip:

Mode switch is important concept.

♦ Chunk: Interview Questions

1. What is OS?

Manages hardware and provides services.

2. Process vs Thread?

Process = independent, Thread = lightweight.

3. What is Deadlock?

Processes waiting forever.

4. Conditions of Deadlock?

4 necessary conditions.

5. What is Scheduling?

CPU allocation process.

6. Best scheduling algorithm?

Depends on use-case (SJF optimal).

7. What is Paging?

Fixed-size memory allocation.

8. Paging vs Segmentation?

Fixed vs variable size.

9. What is Virtual Memory?

Disk used as RAM.

10. What is IPC?

Process communication methods.

11. What is Kernel?

Core OS component.

12. System call example?

read(), write()

13. What is context switching?

Switching CPU between processes.

14. What is fragmentation?

Wasted memory space.

15. Internal vs External fragmentation?

Inside vs outside blocks.

16. What is thrashing?

Excessive paging.

17. What is semaphore?

Synchronization tool.

18. What is mutex?

Mutual exclusion lock.

19. What is file system?

Data organization method.

20. What is RTOS?

Time-critical OS.